

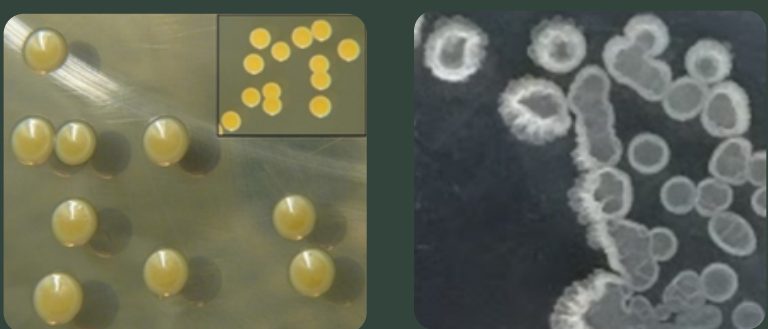
POLYPHENOL EXTRACTION FROM CASHEW TESTA AND ITS APPLICATION IN SKIN CREAM FORMULATION

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III. Material



Cashew testa from Binh Phuoc Province



Staphylococcus,
Candida albicans



Folin-ciocalteu, NaOH, FeCl₃, DPPH, ABTS, ...

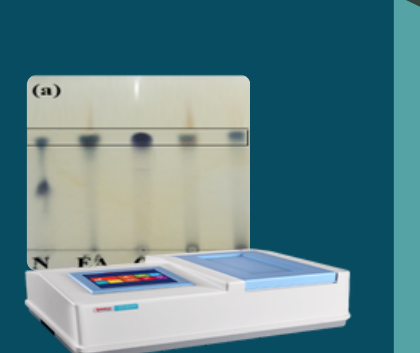
III. Experimental Process



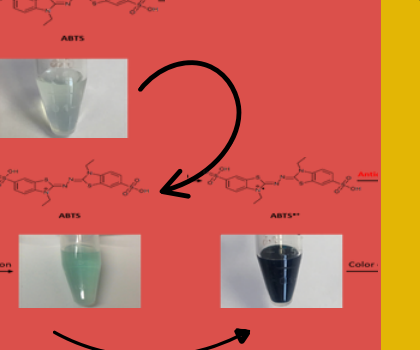
Raw Material Preparation



Ultrasound-Assisted Extraction (UAE)



Determination of polyphenol content



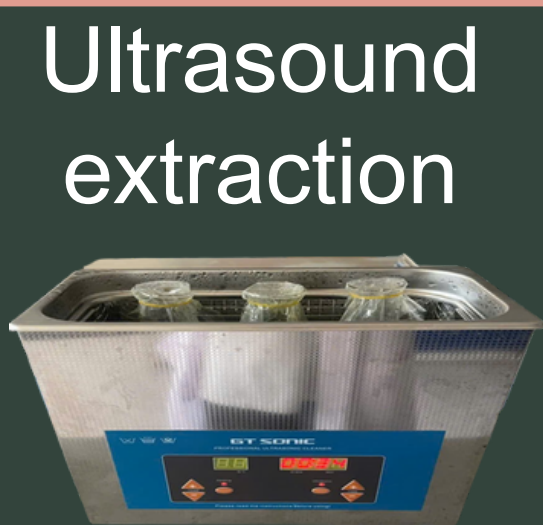
Biological Activity
Antioxidant
Antibacterial



Packaging Product

IV.Extraction Process

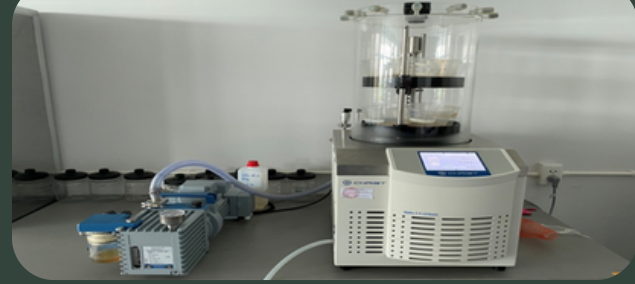
Cashew Testa Powder Sample



Collect Dried Extract Powder

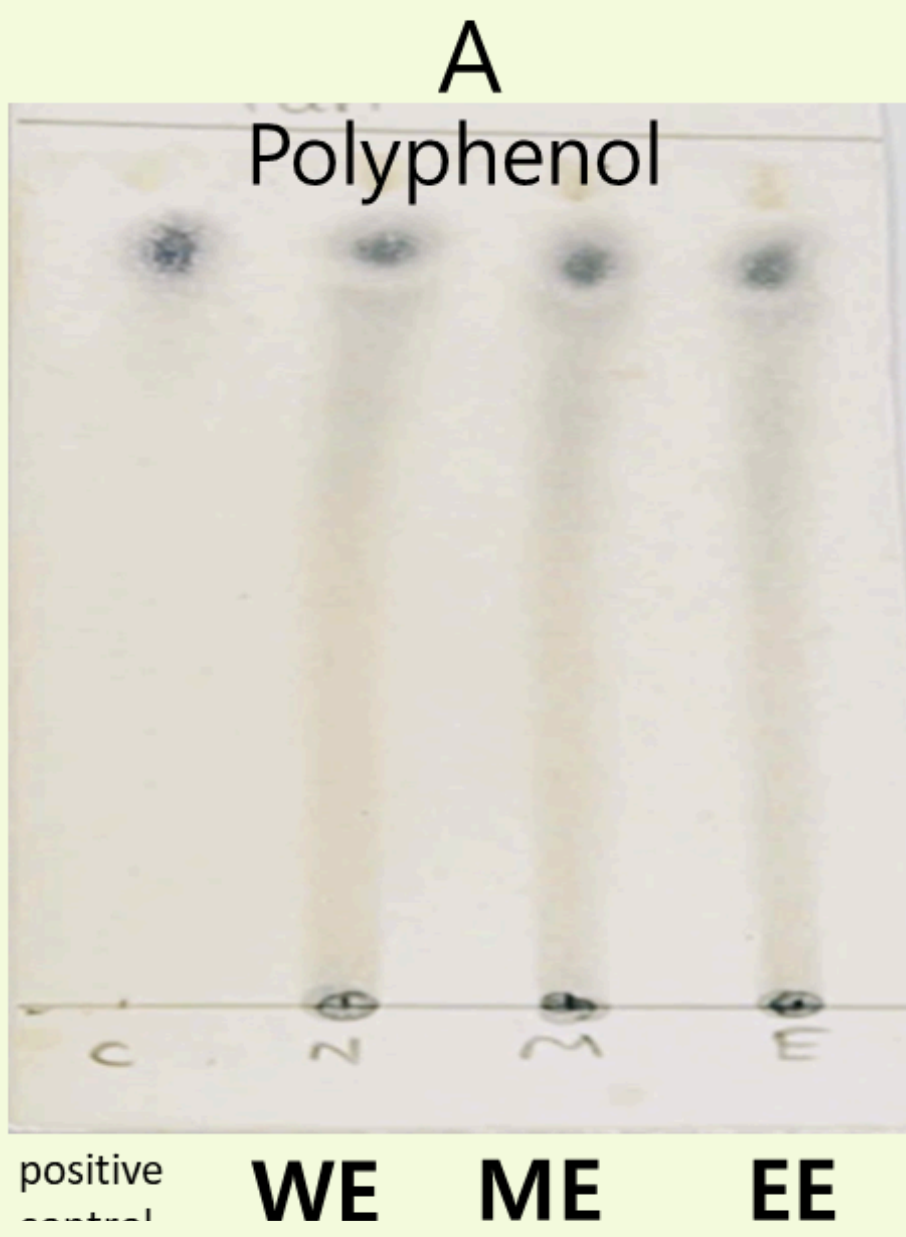
Freeze-drying of Extract Solution

Rotary Evaporator to Remove Solvent

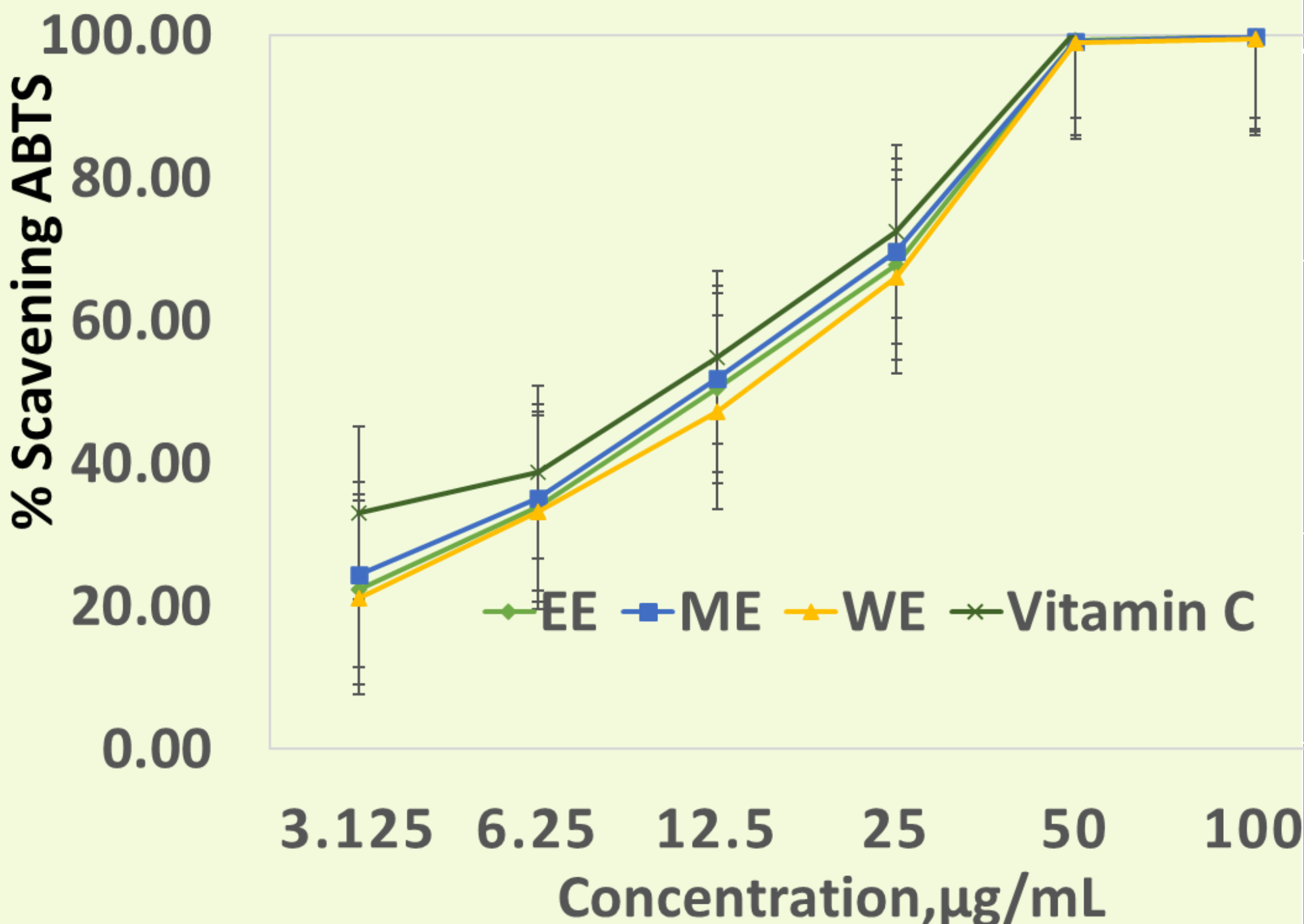


V. Result

1. Phytochemical Screening and Polyphenol Content



3. Antioxidant activity

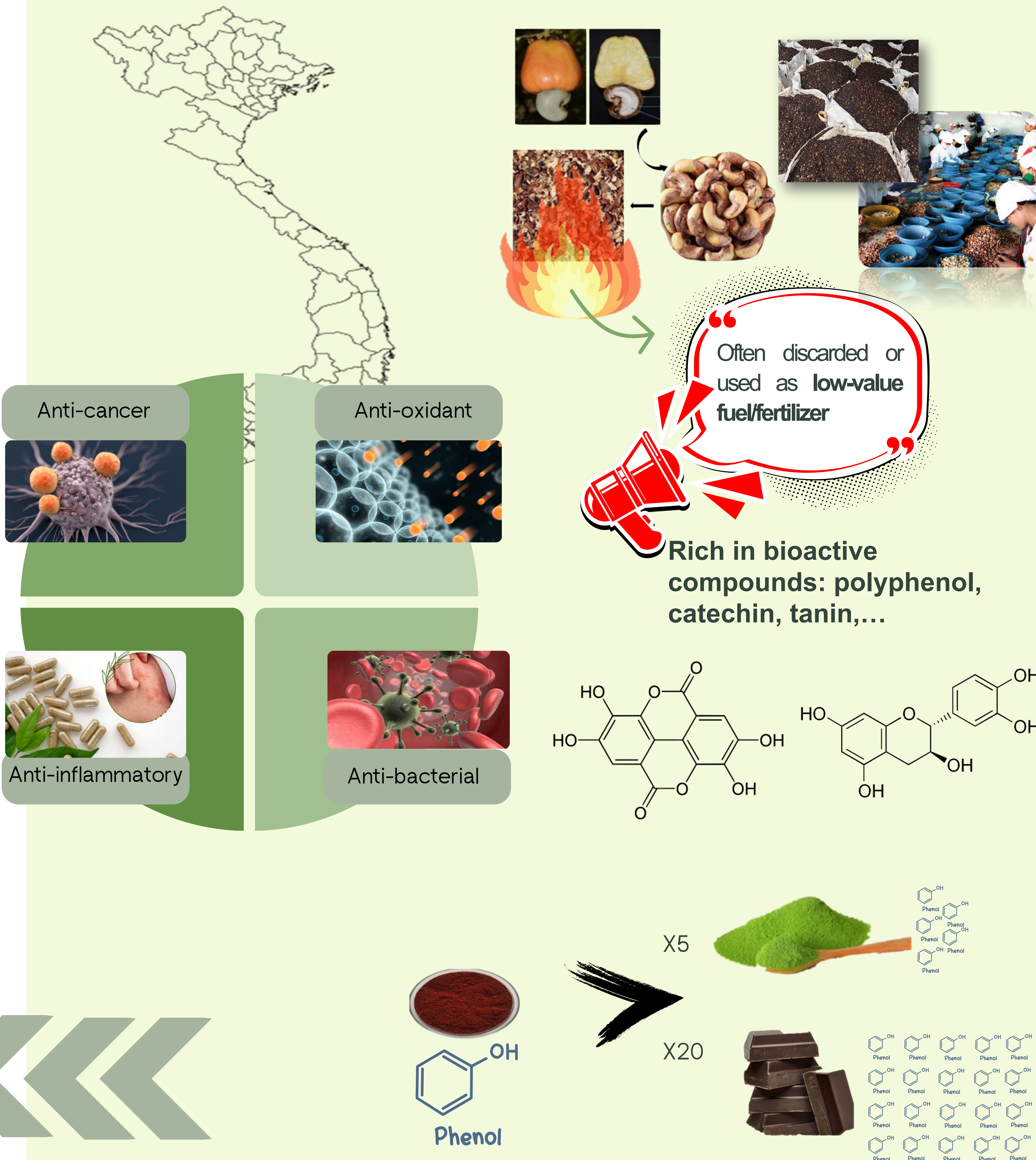


2. Total Polyphenol Content

Sample	Extraction efficiency (%)	Total polyphenol content GAE (mg/g extract)
EE	26.53± 2.87	485.33± 4.26
ME	36.07± 1.7	491.15±4.69
WE	15.47± 0,31	453.13±0.26

Sample	Equation	IC50 value
EE	y= 24.282ln(x) - 6.9244 R2= 0.9545	10.43
ME	Y = 23.646ln(x) - 4.7629 R2= 0.9462	10.14
WE	y= 24.956ln(x) - 10.874 R2= 0.9526	11.36
Vitamin C	y=22,245ln(x) + 2,8521 R2= 0.9535	8.32

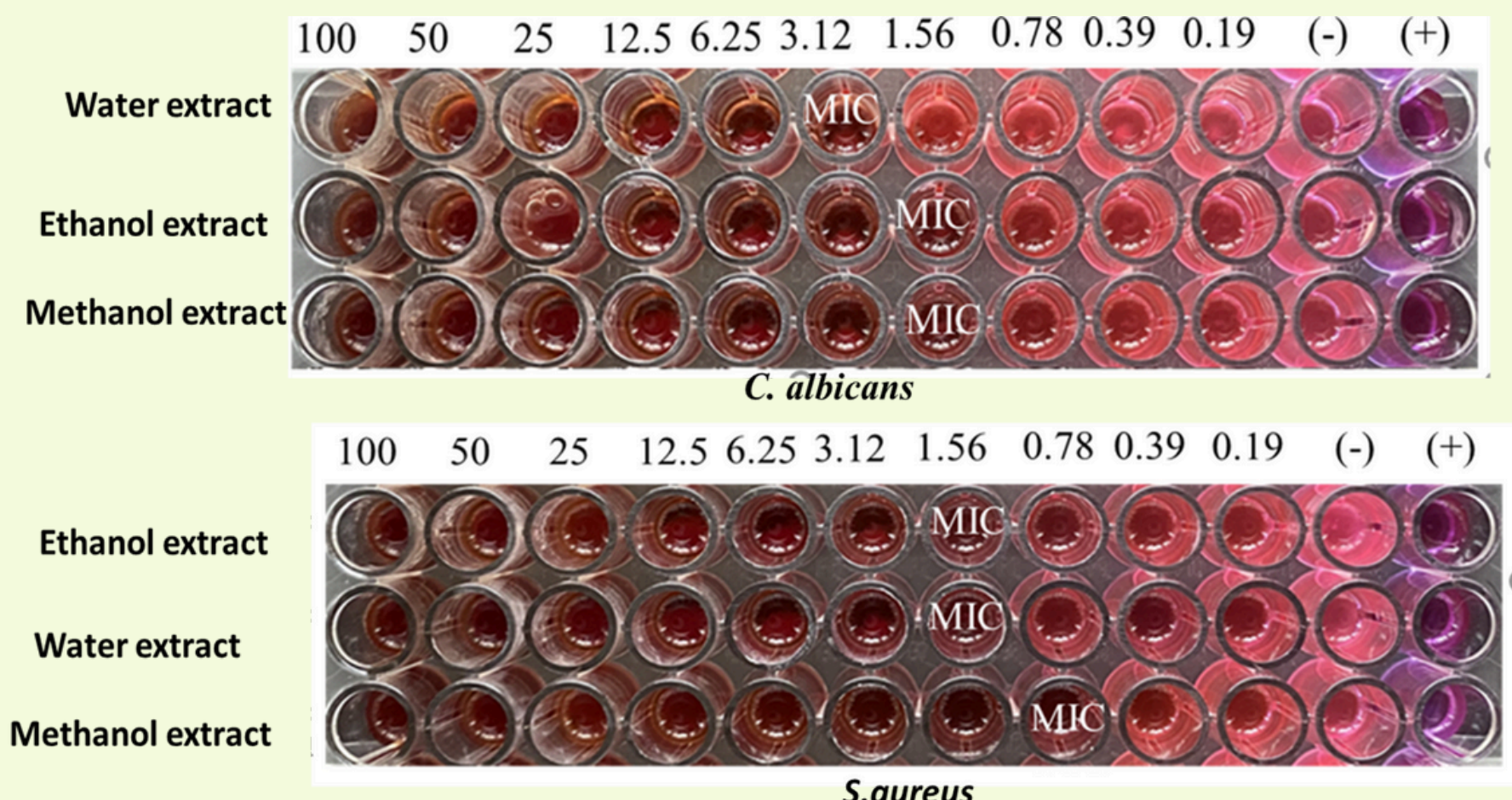
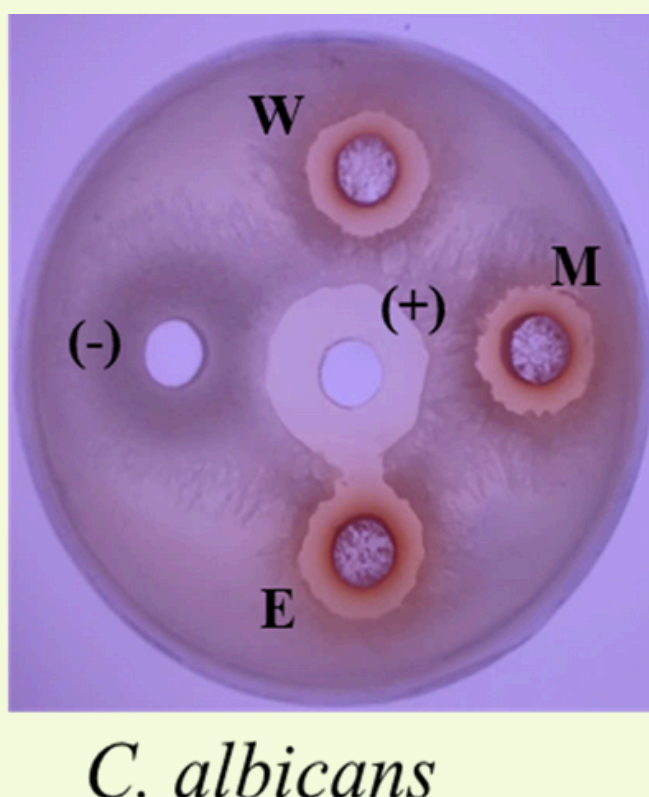
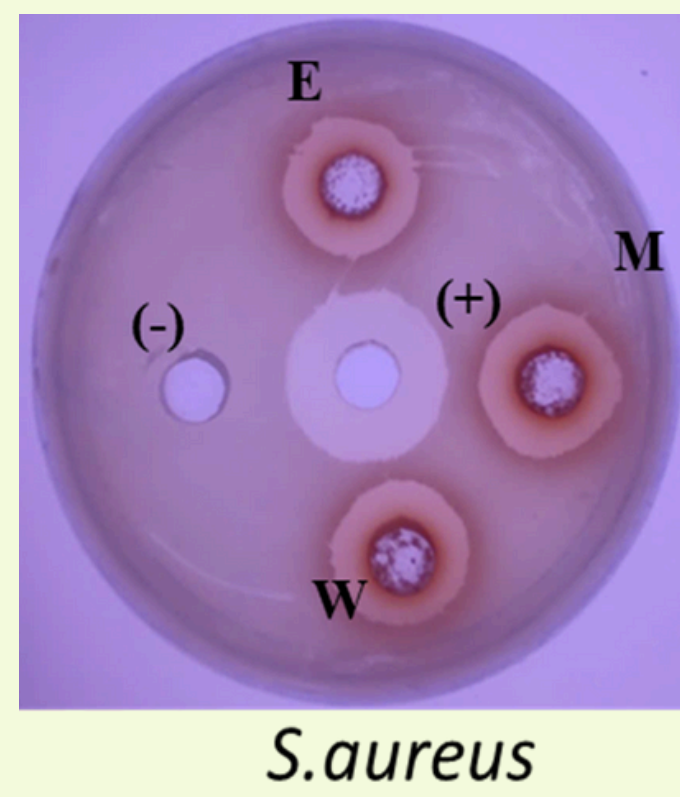
I. Introduction



II. Objective

The objective of this study is to sustainably valorize cashew testa by extracting its bioactive polyphenols and developing an eco-friendly antioxidant cream, thereby promoting green innovation, functional product development, and responsible agricultural waste utilization

4. Antibacterial Activity



VI. Conclusion

- Cashew testa extract (especially ethanol-based) exhibited high polyphenol content with strong antioxidant and antimicrobial activities.
- A stable, natural cream was successfully formulated using the extract, demonstrating practical cosmetic application.
- The project promotes waste valorization and supports sustainable, eco-friendly innovation in agriculture and skincare.

VII. Acknowledgement

We thank our mentors from our heart, Biotechnology center of HCM City laboratories, and all supporters for enabling this project.